CS 489/689-1001: Fundamentals of Deep Learning
Assignment #4 Report
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A. Overview

This report describes the implementation of a binary segmentation deep learning model for skin lesions. Within this report, three model variants will be observed and analyzed: U-Net trained from scratch, ResUNet with a frozen ResNet encoder, and a completely fine-tuned ResUNet. Model performance, loss functions, and a visual comparison will all be compared within this report.

B. Quantitative Comparison: Model Variants

*** **Table 1** - Comparison of the three segmentation model variants. Higher is better for ACC, Dice, and IoU. Lower is better for FP% and FN%. (Dice loss)*

Model Variant	Acc ↑	Dice ↑	IoU ↑	FP% ↓	FN% ↓
U-Net (from scratch)	0.9054	0.8339	0.7161	0.0750	0.1402
ResUNet (encoder frozen)	0.9220	0.8633	0.7605	0.0630	0.1156
ResUNet (fine-tune all)	0.9431	0.8954	0.8118	0.0338	0.1841

C. Quantitative Comparison: Loss Functions

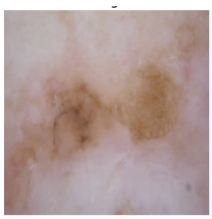
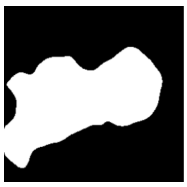
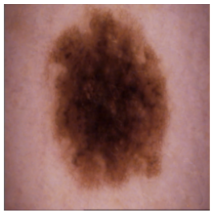
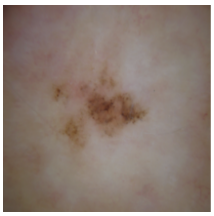
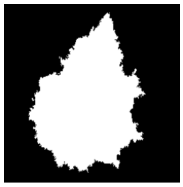
*** **Table 2** - Comparison of Dice and Dice+BCE losses using ResUNet (fine-tune all). Higher is better for ACC, Dice, and IoU. Lower is better for FP% and FN%*

Loss Function	Acc ↑	Dice ↑	IoU ↑	FP% ↓	FN% ↓
Dice	0.9431	0.8954	0.8118	0.0338	0.1140
Dice + BCE	0.9378	0.8895	0.8021	0.0463	0.0100

D. Visual Comparison

*** **Figure 1** -Visual comparison grid. Rows: different test images. Columns: input image, ground-truth mask, and predictions from the three required model variants. (Dice loss)*

Image Name	Input Image	Mask	UNet_Scratch	ResNet_Frozen	ResNet_Pretrained

ISIC_0022006					
ISIC_0018580					
ISIC_0036070					
ISIC_0036312					

E. Short Questions

1. **Did pretraining improve segmentation performance compared to training from scratch?**
   Pretraining improved segmentation performance compared to scratch, as shown in the metrics table. All metrics (Acc, Dice, IoU, FPR, and FNR) were worse in the from-scratch model compared to the other two.

2. **How did freezing the encoder compare to fine-tuning all layers?**
   From the results computed, freezing the encoder performed worse in Acc, Dice, IoU, and FPR compared to fine-tuning all layers. However, the frozen encoder did perform much better in FNR. Given the context of skin lesion detection, a low false-negative rate can be argued to be the most important metric.

3. **Did Dice+BCE improve the results compared to Dice alone?**
   Just like the freeze-encoder vs. fine-tuning-all experiments above, Dice + BCE performed worse Acc, Dice, IoU, and FPR compared to Dice alone. However, Dice + BCE did perform much

better in FNR. Given the context of skin lesion detection, a low false-negative rate can be argued to be the most important metric.

4. **Which model produced the best visual masks?**
   After visually observing the image comparison grid, there is no universal model that objectively produces the best visual masks for all images. Some models overfit or underfit depending on the input image.

   For ISIC_0022006, I argue ResNet_Pretrained performed the best.
   For ISIC_0018580, I argue UNet_Scratch and ResNet_Frozen performed equally well.
   For ISIC_0036070, I argue UNet_Scratch performed the best.
   For ISIC_0036312, I argue ResNet_Frozen performed the best.

   Due to this, UNet_Scratch and ResNet_Pretrained perform on these four cases alone. However, it is important to note that this sample size is small (only 4), and most definitely doesn't capture the performance adequately for all cases.

F. Conclusion

This assignment has highlighted the drastic importance of transfer learning and loss decisions. With deep learning architectures now being leveraged in medical applications, as shown with skin lesion segmentation, it is imperative to optimize models as best as possible.

Transfer learning, in both of the cases tested, showcases the importance of leveraging already-known weights and adapting to newfound situations. Aside from medical instances, this pipeline can be applied universally, optimally, efficiently, and quickly to develop models without as many computational constraints.

In the case of our experiments for binary skin lesion segmentation, overall, it can be argued that the **frozen-encoder** and **Dice + BCE Loss** performed the best given the context. Although lacking in other metrics, false negatives are certainly the most imperative metric. It is morally worse to misdiagnose a melanoma lesion as benign than to misdiagnose a benign lesion as melanoma. Consequently, these experiments demonstrate that metrics should not be equally prioritized.